INCH-POUND

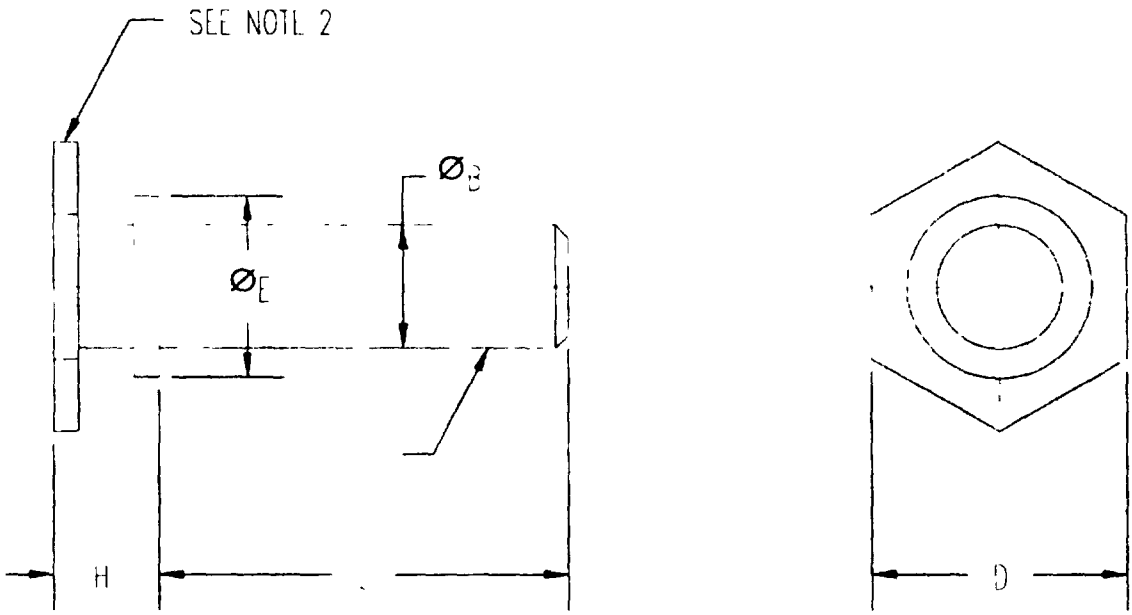
MIL-S-63540/3A
21 June 1995
SUPERSEDING
MIL-S-63540/3
31 March 1986

MILITARY SPECIFICATION SHEET

STUD, SELF-CLINCHING, FLUSH HEAD, HEX CLINCH TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the studs described herein shall consist of this specification and the latest issue of MIL-S-63540.



NOTES

1. All dimensions are in inches.
2. Clinching profile may vary.
3. Head relief diameter optional.

FIGURE 1 STUD, SELF-CLINCHING, FLUSH HEAD, HEX CLINCH TYPE.

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited.

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TABLE I PART DIMENSIONS

DASH NO	T THREAD SIZE CLASS 2A	ØB STUD DIA	D +/- 015 HEX ACROSS FLATS	ØE	H +/- 003 HEAD HEIGHT	SHEET THICKNESS MIN (REF)
1					038	040
2	086-56 UNC (2-56)	086	187	157	058	062
3					085	093
4					117	125
5					038	040
6	112-40 UNC (4-40)	112	187	157	058	062
7					085	093
8					117	.125
9					038	040
10	112-40 UNC (4-40)	112	250	199	058	062
11					085	093
12					117	125
13					038	040
14					058	062
15	138-32 UNC (6-32)	138	250	199	085	093
16					.117	.125
17					147	.156
18					177	187
19					038	.040
20					058	.062
21	164-32 UNC (8-32)	.164	.312	.261	085	093
22					.117	125
23					.147	156
24					177	.187
25					038	040
26					.058	.062
27	190-32 UNF (10-32)	190	.312	261	085	093
28					117	125
29					147	156
30					177	.187
31					117	125
32					.147	156
33	250-20 UNC (1/4-20	250	437	.382	177	187
34					236	.250
35					296	.312
36					117	.125
37					.147	.156
38	250-28 UNF (1/4-28	250	437	382	177	187
39					236	250
40					296	312

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REQUIREMENTS

1 Material Corrosion resistant steel shall be Type (UNS S30430) (formerly identified as XM-7) and 302 (UNS S30200) in accordance with the chemical compositions specified in ASTM A 493 or Types 303 (UNS S30300) and 303Se (UNS S30323) in accordance with ASTM A 582.

2 Surface treatment Corrosion resistant steel studs shall be cleaned, descaled and passivated in accordance with QQ-P-35.

3. Hardness Corrosion resistant steel studs shall have a hardness of 171BHN minimum

4. Push-out values When the stud is properly installed, (see installation procedures), push-out values shall be as specified in Table II

TABLE II PUSH-OUT VALUES

HEX SIZE NOMINAL	SHEET THICKNESS MIN 1/ (REF)	PUSH-OUT LOAD (LBS) MIN 2/	PUSH-OUT LOAD (LBS) MIN 2/
		ALUMINUM ALLOY SHEET MATERIAL 3/	COLD ROLLED STEEL SHEET MATERIAL 3/
.187	.040	100	125
	062	210	250
	.093	300	325
	.125	350	500
.250	040	150	150
	.062	275	300
	.093	325	350
	125	450	650
312	.040	185	200
	.062	330	350
	.093	350	425
	.125	550	900
.437	.062	395	400
	.093	490	500
	.125	900	1,000

1 Any sheet thickness above the minimum indicated for a specific stud may be used.
2 The load values shown are the minimum non-destruct values that apply during testing.
3 Sheet material hardness values shall not exceed 70HRB

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5 Part number The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table I, followed by a dash number denoting stud length in increments of .0625

Example M63540/3-4-6

- M63540/3 — Specification sheet number
- -4 — Dash number
- -6 — Length in increments of .0625.

MS3540/3-4-6 indicates - Stud, self-clinching, flush head, hex clinch type, CRES, 086-56 UNC-2A thread, .375 length, .117 head height, .187 hex dimension

INSTALLATION PROCEDURES

a Drill or punch hole in sheet material with diameter as shown in Table III If hole is punched, stud should enter from punch exit side Do not deburr

b Locate head of stud squarely in the drilled or punched hole in the mounting surface with anvil below sheet material (see Figure 2).

c Apply force using a flat faced punch in an arbor press of sufficient capacity, either hand or powered, until the head of stud is embedded flush into sheet material (see Figure 2).

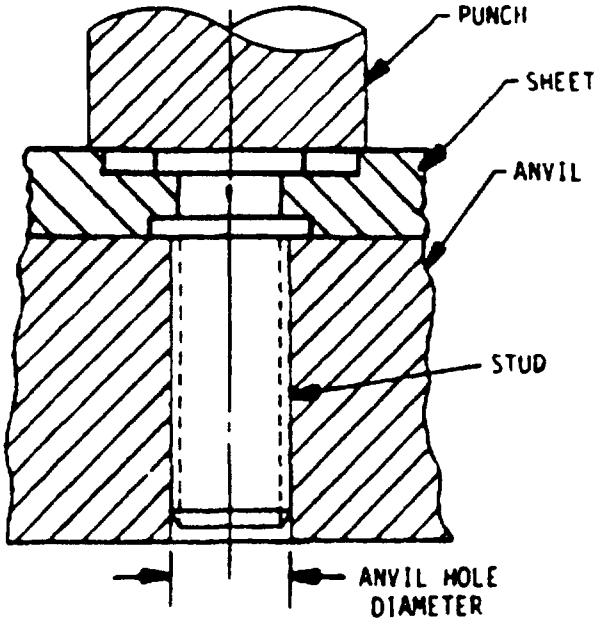


FIGURE 2. GENERAL TOOLING SETUP

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TABLE III INSTALLATION DIMENSIONS

DASH NO	HOLE SIZE IN SHEET ±.002 (REF)	CENTER OF HOLE TO SHEET EDGE MIN (REF)	NUT TIGHTENING TORQUE (IN LBS) MAX (REF)	ANVIL HOLE DIAMETER + 500 - 000 (REF)
1 2 3 4	161	240	2	088
5 6 7 8	.161	240	5	.115
9 10 11 12	.203	305	5	115
13 14 15 16 17 18	.203	305	9	140
19 20 21 22 23 24	.265	400	18	167
25 26 27 28 29 30	.265	400	32	.195
31 32 33 34 35	.386	700	80	.255
36 37 38 39 40	386	700	80	.255

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Custodians
Army - AR
Navy - MC
Air Force - 99

Review activities
Army - AT, ER, ME
Navy - SH, OS
Air Force - 11, 82

NSA - NS

Preparing activity
DLA - IS

(Project 5307-0576)

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